



CSCI 446

ARTIFICIAL INTELLIGENCE

Lecture: 2

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INTELLIGENT AGENTS

In which we discuss the nature of agents, the diversity of environments, and various agent types.

WHAT IS AN AGENT?

- ⚡ **Definition:** An agent perceives its environment via **sensors** and acts upon it through **actuators**.
- ⚡ **Observation:** Some systems, like flush toilets and calculators, seem like agents but are actually artifacts.

AGENTS VS. ARTIFACTS

- ✂ **Artifacts:** Devices like flush toilets perform actions based on simple rules, but do not adapt or learn.
- ✂ **Example:** Calculator reacts deterministically to inputs without decision-making capability.
- ✂ **Insight:** Important to distinguish between true agents and simple reactive systems.

RATIONAL AGENTS

- ✂ A rational agent does the "right thing."
- ✂ **Performance Measure:** Key to evaluating desirability of actions but can be challenging to define correctly.

IMPORTANCE OF ENVIRONMENT

- ✂ **Task Environment:** Consists of the performance measure, environment, actuators, and sensors (PEAS).
- ✂ Types:
 - ✂ Fully vs. partially observable
 - ✂ Single-agent vs. multiagent
 - ✂ Deterministic vs. nondeterministic

TYPES OF AGENTS

Simple Reflex Agents

- ✂ **Behavior:** Act based on current percept only.
- ✂ **Example:** Basic vacuum agent.

Model-based Reflex Agents

- ✂ **Behavior:** Maintain internal state based on history of percepts.

Goal-based Agents

- ✂ **Behavior:** Actions achieve specified goals.

Utility-based Agents

- ✂ **Behavior:** Actions maximize perceived utility.

LEARNING AGENTS

- ✂ Capabilities include learning from interactions to improve performance over time.
- ✂ **Components:**
 - ✂ **Performance Element**
 - ✂ **Learning Element**
 - ✂ **Critic**
 - ✂ **Problem Generator**

Agent Design Considerations

- ✂ **Artifacts vs. Agents:** Distinguish design based on adaptability and decision-making capabilities.
- ✂ **Scalability and Flexibility:** Focus on building systems that can learn from experience and adjust accordingly.

DESIGN CONSIDERATIONS

- ✂ **Scalability and Flexibility:** Construct designs that can adapt to different and complex environments.
- ✂ Emphasize the need for agents to learn and adapt autonomously over time.

SUMMARY

- ✂ Understanding agents requires considering their capabilities, environment, and autonomy.
- ✂ Real-life applications necessitate a balance between predefined rules and adaptability.

CONCLUSION

This chapter establishes a foundational understanding of intelligent agents, exploring their design, environment considerations, and types, as essential components of AI systems.

INTEGRATED DEVELOPMENT ENVIRONMENT



THANK

You